

Ruichen XU (徐睿晨)

Postdoctoral Researcher, University of Vienna

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Current Place of Residence: Vienna, Austria

Employment

- **University of Vienna** *Postdoctoral Researcher* Austria, Sept 2026 - Present
 - Mentor: Harald GROBNER, cosupervised by Adel BETINA,
 - Under the FWF Stand-Alone Grant project *Eigenvarieties and p -adic L -functions* (P 325085-N35).

Education

- **Chinese Academy of Sciences** *Ph.D student in mathematics* China, Sept 2021 - Jun 2026
 - Doctoral Advisor: Xin WAN.
 - Doctoral Thesis: *Iwasawa Theory for Unitary Groups via Eisenstein Congruences and its Applications*.
- **Beihang University** *Bachelor of Science in mathematics* China, Sept 2017 - July 2021
 - Undergraduate Advisor: Peng GAO and Shijin ZHANG (thesis advisor).
 - Undergraduate Thesis: *The Proof of Fermat's Last Theorem: Wiles' Proof and Beyond*

Publications and Preprints

- ✓ **On the Mordell-Weil rank of certain CM abelian varieties over anticyclotomic towers.**
(with Haidong LI) *Journal of Number Theory* 293 (2027) pp. 1-41.
DOI: [10.1016/j.jnt.2026.06.001](https://doi.org/10.1016/j.jnt.2026.06.001) (open access). arXiv: [2502.12648](https://arxiv.org/abs/2502.12648).
- **The Gan-Gross-Prasad period of Klingen Eisenstein families over unitary groups**
Preprint (2024).
arXiv: [2410.13132](https://arxiv.org/abs/2410.13132).

Visiting Experiences

- **Université Marie et Louis Pasteur** *Visiting Doctoral Student* France, Dec 2024 - July 2025
 - Host Advisor: Olivier FOUQUET.
 - Funded by the program “2024 International Research Collaboration Program for Postgraduate Students” of Chinese Academy of Sciences (1 350 EUR monthly).
- **Peking University** *13th Enhanced Program for Graduate Study* China, Spring 2021
- **Peking University** *12th Enhanced Program for Graduate Study (Online)* China, Spring 2020

Conference Talks

- **33èmes Journées Arithmétiques** July 1, 2025
Title: The Gan-Gross-Prasad period of Klingen Eisenstein families over unitary groups.
Location: Université du Luxembourg. Esch-sur-Alzette, Luxembourg.
- **Géométrie algébrique et théorie des nombres en Bourgogne Franche-Comté** Mar 28, 2025
Title: The method of Eisenstein congruences in Iwasawa theory.
Location: Institut de Mathématiques de Bourgogne. Dijon, France.

Teaching Experiences

- **Advanced Abstract Algebra** *Teaching Assistant, at Beihang University* Fall 2020
Undergraduate course on Galois theory, the theory of modules and the representation theory.
- **Complex Analysis** *Teaching Assistant (Online), at Beihang University* Spring 2020
Undergraduate course on basic complex analysis for students majoring in mathematics.
- **Undergraduate Mentors Program** *Mentor, at Beihang University* Fall 2018 - Spring 2019
Guiding freshmen in making a smooth transition from middle school to undergraduate studies by providing psychological and administrative support, along with academic guidance in foundational courses including calculus, mathematical analysis, linear algebra, programming (C/Python), and physics.

Scholarships

- **Excellent Award of the Dean's Scholarship** *AMSS, CAS* 2025
for outstanding graduating students at the Academy of Mathematics and Systems Science (AMSS), Chinese Academy of Sciences (CAS). (22 000 CNY).
- **Professor Hua Loo-Keng Scholarship** *AMSS, CAS* 2023, 2024
for excellence in PhD qualifying exams and research (100 000 CNY each).
- **Professor Hua Loo-Keng Scholarship** *AMSS, CAS* 2019, 2020, 2021
for excellence in undergraduate mathematical coursework and contests (2 000 CNY each).

Conferences Attended

- **2026 Jinling Conference of Number Theory** *Nanjing, Jiangsu, China* Apr 2026
At Nanjing University.
- **Relative Langlands and Arithmetic** *Marseille, France* Feb 2026
At the Centre International de Rencontres Mathématiques.
- **Conference in honor of Professor George Lusztig** *Hong Kong SAR, China* Nov 2025
At the University of Hong Kong.
- **33èmes Journées Arithmétiques** *Esch-sur-Alzette, Luxembourg* July 2025
At the Université du Luxembourg.
- **Iwasawa Theory and Representations of p -Adic Groups** *Lille, France* June 2025
At the Laboratoire Paul Painlevé, Université de Lille.
- **Géométrie algébrique et théorie des nombres en Bourgogne Franche-Comté** Mar 2025
At the Institut de Mathématiques de Bourgogne, Dijon, France.
- **Young researchers in Galois Representations and related topics** *Genova, Italy* Jan 2025
At the Università di Genova.
- **Workshop on p -adic Geometry** *Singapore* Nov 2024
At the National University of Singapore.
- **BSD Conjecture and Goldfeld Conjecture** *Kunming, Yunnan, China* Oct 2024
At Tianyuan Mathematics Research Center.
- **Development of Iwasawa Theory** *Yokohama, Kanagawa, Japan* Jul 2024
A conference in honor of the 60th birthday of Masato Kurihara. At Keio University.

- **2024 Jinling Conference of Number Theory** *Nanjing, Jiangsu, China* Apr 2024
At *Nanjing University*.
- **Special Values of L -functions** *Sanya, Hainan, China* Jan 2024
At *Tsinghua Sanya International Mathematics Forum*.
- **The 3rd Cross Strait Workshop on Arithmetic Geometry** *Hefei, Anhui, China* Aug 2023
At the *University of Science and Technology of China*.
- **Workshop on Arithmetic and Topology** *Shenzhen, Guangdong, China* Aug 2023
At the *Southern University of Science and Technology*.
- **Iwasawa 2023: in memory of John Coates** *Cambridge, the United Kingdom* July 2023
At the *University of Cambridge*.
- **Satellite Conference in Number Theory of ICBS** *Beijing, China* July 2023
At *Morningside Center of Mathematics, Chinese Academy of Sciences*.
- **2023 Jinling Conference of Number Theory** *Nanjing, Jiangsu, China* June 2023
At *Nanjing University*.

Summer Schools Attended

- **Summer school on Galois representations and Reciprocity** *Hanoi, Vietnam* July 2024
At *Vietnam Institute for Advanced Study in Mathematics*.
- **2021 Summer School on Number Theory** *Xi'an, Shaanxi, China* July 2021
At *Xi'an Jiaotong University*.
- **Theta Correspondence and Its Arithmetic Applications** *Suzhou, Jiangsu, China* July 2021
At *Soochow University*.
- **Short Courses on Number Theory** *Xiamen, Fujian, China (Online)* July 2020
At *Xiamen University*.
- **2020 SCMS Algebraic Geometry Summer School** *Shanghai, China (Online)* July 2020
At *Shanghai Center of Mathematical Sciences, Fudan University*.
- **"The Road to Mathematics" Int'l Summer School** *Harbin, Heilongjiang, China* July 2019
At *Harbin Institute of Technology*.

Seminars Organized and Attended

As the organizer and the main speaker of the following seminars among students:

- **Universal Iwasawa theory** Fall 2024
Study the the [article](#) "*The Iwasawa Main Conjecture for universal families of modular motives*" by O.Fouquet and X.Wan.
- **Automorphic methods in Iwasawa theory** Spring 2023 - Fall 2023, Spring 2025
Study the machinery of Eisenstein congruences via the [article](#) "*The Iwasawa Main Conjectures for GL_2* " by C.Skinner and E.Urban, and its later developments.
- **Iwasawa Theory of elliptic curves** Fall 2021, Fall 2023
Study the [article](#) "*Introduction to Iwasawa Theory for Elliptic Curves*" by R.Greenberg, with a special focus on solving all the exercises.

- **Automorphic Forms and Automorphic Representations** Spring 2022 - Fall 2022
Following the **textbook** “*An Introduction to Automorphic Representations*” by J.Getz and H.Hahn.

As the organizer and a speaker of the following seminars among students:

- **Rational points on elliptic curves** Fall 2024
Following the **textbook** “*Rational Points on Modular Elliptic Curves*” by H.Darmon.
- **Arithmetic of elliptic curves** Fall 2023 - Spring 2024
Following the textbooks “*Arithmetic of Elliptic Curves*” and “*Advanced Topics in the Arithmetic of Elliptic Curves*” by J.Silverman.

Languages

- **Chinese:** Native
- **English:** Proficient
- **French:** Beginner